CYCLISTIC BIKE-SHARE

How Does a Bike-Share Navigate Speedy Success?

A Data Analysis Case Study — Converting Casual Riders into Annual Members

Prepared by: Prateek Gupta

Prepared for: Lily Moreno, Director of Marketing Data Source: Divvy / Cyclistic Historical Trip Data — 2022, 2023 & 2024 (Tableau Visualizations) (Profile - prateek.gupta1807 | Tableau Public)

Framework: Ask — Prepare — Process — Analyze — Share — Act

---

1. Ask — The Business Task

Cyclistic is a Chicago-based bike-share program with 5,824 bicycles across 692 stations, offering single-ride passes, full-day passes, and annual memberships. The finance team has determined that annual members are significantly more profitable than casual riders, and the Director of Marketing, Lily Moreno, has set a strategic goal of converting casual riders into annual members rather than acquiring entirely new customers.

Business Task Analyze historical Cyclistic trip data to answer the guiding question: How do annual members and casual riders use Cyclistic bikes differently? The resulting insights will inform a marketing strategy designed to convert casual riders into annual members.

Key Stakeholders

- Lily Moreno — Director of Marketing (primary stakeholder, approves strategy)
- Cyclistic Marketing Analytics Team — designs and executes the campaign
- Cyclistic Executive Team — approves the final recommendations

Deliverable: A clear, data-backed statement of how usage patterns differ between casual riders and annual members, translated into three actionable marketing recommendations.

---

2. Prepare — Data Sources

The analysis uses Cyclistic's (Divvy's) historical bike trip data, made publicly available by Motivate International Inc. under an open data license. Three years of ride-level data were reviewed and visualized: 2022, 2023, and 2024. Each record includes ride ID, bike type, start/end timestamps and stations, and a rider-type field distinguishing casual riders from annual members.

Data Credibility (ROCCC Check)

Criteria	Assessment
Reliable	Data is collected directly from Cyclistic's own docking/GPS systems — not survey-based, so it is objective and complete for the fields it tracks.
Original	Sourced directly from Motivate International Inc., the operator of the bike-share system.
Comprehensive	Covers every recorded ride across all stations and both rider types for a full 12-month cycle, across three separate years.
Current	Data spans 2022–2024, including the most recent full year, so trends reflect current rider behavior.
Cited	Publicly licensed and attributable to Motivate International Inc.; no proprietary or personally identifiable information is included.

Known Limitations

- Privacy restrictions prevent linking rides to individual riders, so it is not possible to tell whether a casual rider took multiple single-ride trips or where riders live.
- The data cannot explain motivation (why a casual rider rides) — only observed behavior (when, how long, how often).
- Station-level and geographic patterns were not part of the visualizations reviewed for this report and are flagged as an opportunity for further analysis.

---

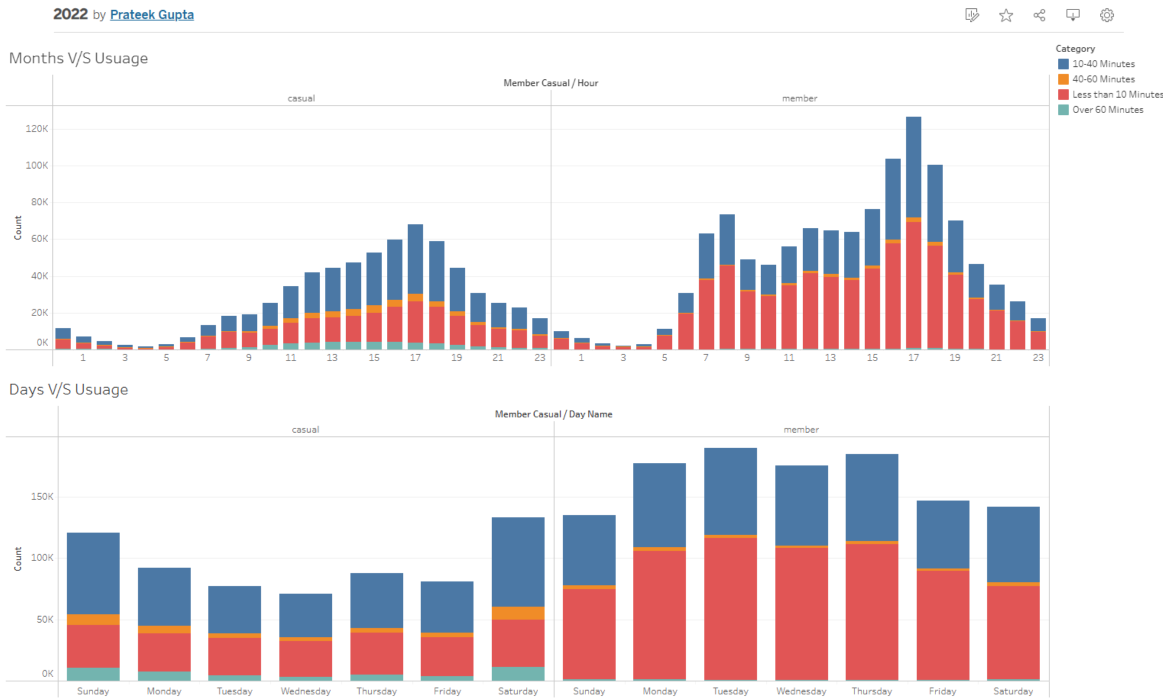
3. Process — Cleaning & Transformation

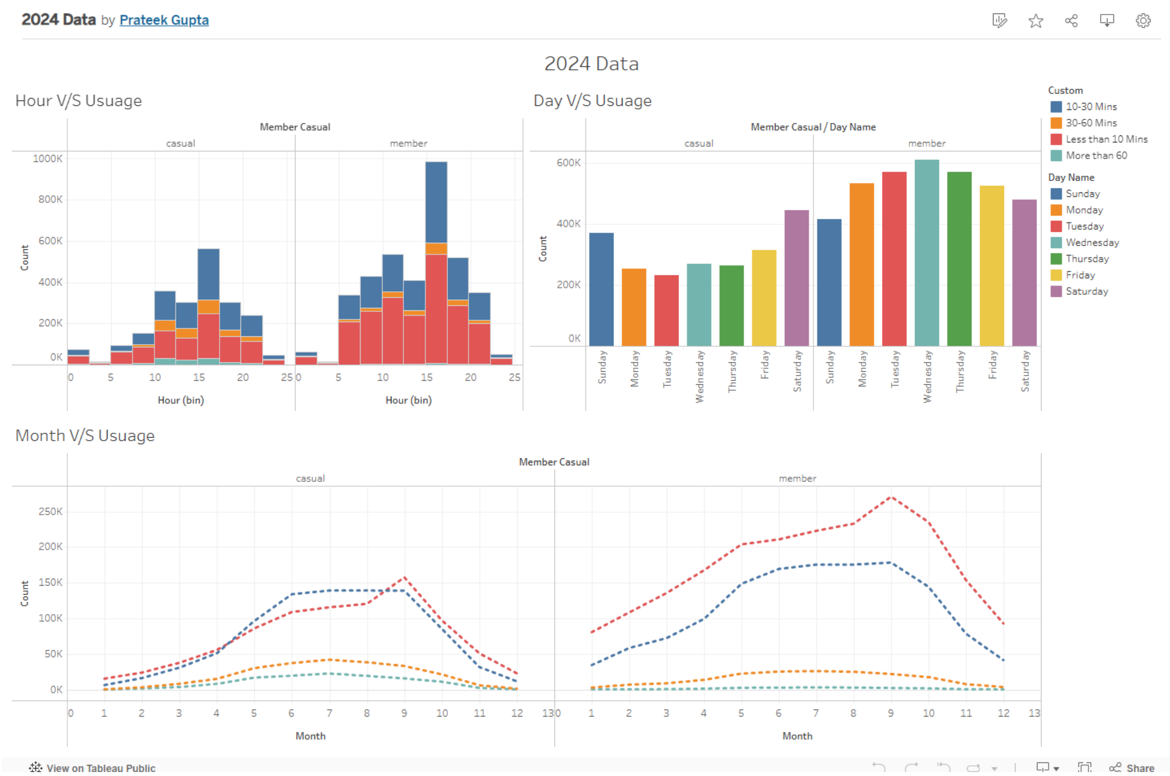
The following steps were applied before analysis, consistent with the case study roadmap:

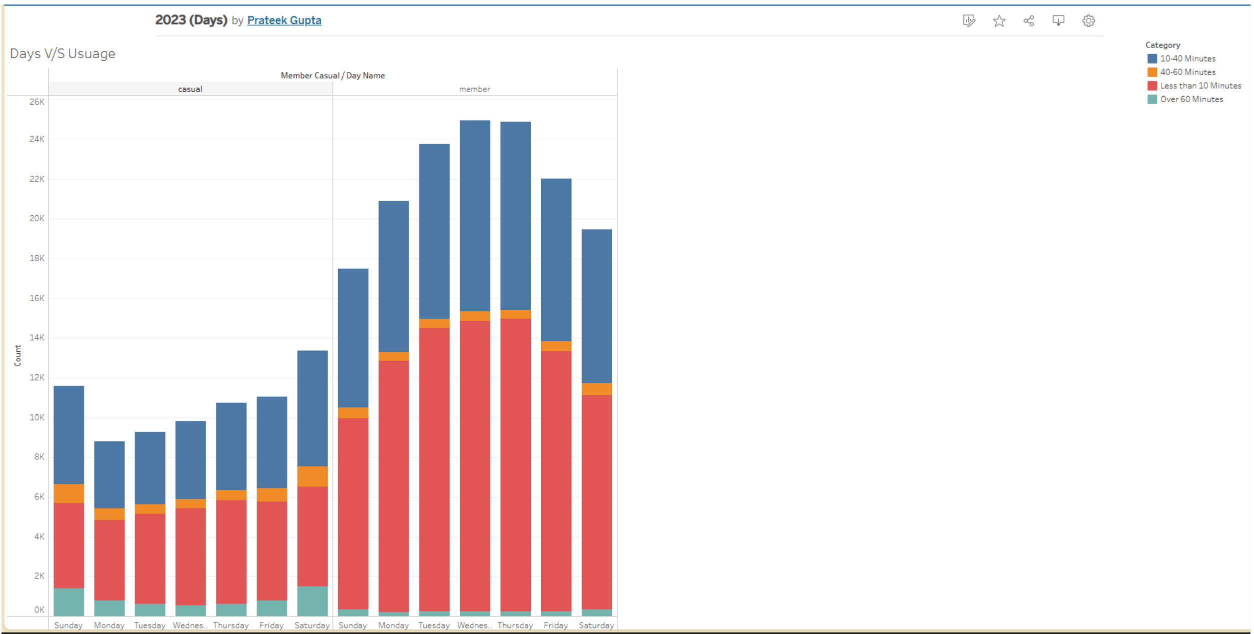
1. Each year's raw trip data (.csv) was collected into year-specific folders (2022, 2023, 2024) to preserve original files.(Index of bucket "divvy-tripdata")

2. A ride_length field was calculated (ended_at − started_at) for every trip and grouped into four duration bins used consistently across all three dashboards: Less than 10 Minutes, 10–30/10-40 Minutes, 30-60/40-60 Minutes, and More than/Over 60 Minutes.
3. A day_of_week field was derived from started_at (1 = Sunday … 7 = Saturday) to allow day-level aggregation.
4. An hour(bin) field was extracted from started_at to allow hour-of-day aggregation.
5. Rows were grouped and filtered by the member_casual field so that every visualization directly compares "casual" vs "member" side by side.
6. Cleaned, aggregated data was loaded into Tableau to build three interactive dashboards: hour-of-day usage, day-of-week usage, and month-of-year usage, each split by rider type and further broken down by ride-duration category.

No personally identifiable information was used at any stage, in line with Cyclistic's data-privacy requirements.


2022 by Prateek Gupta
Months V/S Usage
Member Casual / Hour
casual
member
Count
Category
10-40 Minutes
40-60 Minutes
Less than 10 Minutes
Over 60 Minutes
Days V/S Usage
Member Casual / Day Name
Sunday
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday


2024 Data by Prateek Gupta
2024 Data
Hour V/S Usage
Member Casual
casual
member
Count
Hour (bin)
Day V/S Usage
Member Casual / Day Name
Custom
10-30 Mins
30-60 Mins
Less than 10 Mins
More than 60
Day Name
Sunday
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Month V/S Usage
Month
View on Tableau Public
Share



[2024_CaseStudy_Google_DAP_Cyclistic | Tableau Public]

[2023_CaseStudy_Google_DAP_Cyclistic | Tableau Public]

[2022_CaseStudy_Google_DAP_Cyclistic | Tableau Public]

---

4. Analyze & Share — Key Findings

Three Tableau dashboards were built covering 2022, 2023, and 2024 trip data. Despite being separate years, the same three behavioral patterns hold consistently, which strengthens confidence that these are structural differences between casual riders and members rather than one-off seasonal noise.

Finding 1: Members ride on a commute schedule; casual riders ride for leisure Looking at rides by hour of day (2022 and 2024 dashboards), casual riders show a single broad peak in the early-to-late afternoon (roughly hour 14–18), consistent with recreational, leisure-time riding. Annual members show a distinctly different, twin-peaked commuter pattern — a rise in the morning around hour 7–8 and a much sharper spike in the late afternoon around hour 17 — the classic signature of people riding to and from work.

Finding 2: Casual usage peaks on weekends; member usage peaks on weekdays The day-of-week charts tell the same story in all three years reviewed (2022, 2023, 2024). Casual ridership climbs steadily through the week and peaks on Saturday, with Sunday also elevated — clear weekend, leisure-oriented behavior. Member ridership is highest Tuesday through

Thursday and noticeably dips on Saturday and Sunday, confirming a Monday-to-Friday, work-commute usage pattern.

Finding 3: Casual rides run longer; member rides are quick, point-to-point trips Across every dashboard, the duration breakdown (color-coded by "Less than 10 Minutes," "10–30/10–40 Minutes," "30–60/40–60 Minutes," and "Over/More than 60 Minutes") shows that members' trips are overwhelmingly in the "Less than 10 Minutes" band — short, efficient, point-to-point trips consistent with commuting. Casual riders, in contrast, show a much larger share of rides in the 10–30, 30–60 and even 60-plus minute bands, consistent with exploratory or recreational rides where duration is part of the experience rather than something to minimize.

Finding 4: Both groups are highly seasonal, but casual ridership swings harder The month-over-month view (2024 dashboard) shows both rider types peaking in the summer months (roughly June through September, with the single highest point around month 9) and falling sharply in winter. Casual ridership falls closer to zero in the off-season, while members retain a comparatively higher base of rides even in shoulder months — suggesting members treat Cyclistic as reliable, weather-tolerant transportation, while casual riders treat it as a fair-weather leisure activity.

Summary of Differences

Dimension	Casual Riders	Annual Members
Time of day	Single afternoon peak (~2–6 PM)	Twin peaks: AM commute (~7–8) & PM commute (~5 PM)
Day of week	Peaks Saturday/Sunday (weekend)	Peaks Tuesday–Thursday (weekday)
Ride duration	Skews longer (10–30, 30–60+ min)	Skews shorter (mostly under 10 min)
Seasonality	Sharp swing — near zero in winter	Sustained — still active in off-season
Likely use case	Leisure / recreation / tourism	Commuting / routine transportation

---

5. Act — Top Three Recommendations

The consistent, three-year pattern points to one central insight: casual riders already use Cyclistic like members do in the summer and on weekdays to some extent — they simply haven't been offered a membership structure that matches their actual riding behavior. The recommendations below meet casual riders where their behavior already is.

1. Launch a weekend-focused membership tier or promotion Since casual riders are concentrated on Saturdays and Sundays, target them directly at docking stations and in-app prompts during weekend rides with a "weekend membership" or a discounted trial annual membership offered specifically after a weekend ride. Emphasize the cost savings of an annual plan versus repeatedly paying single-ride or full-day passes for the same weekend habit they already have.

2. Run a peak-season membership push (June–September) Casual ridership is heavily concentrated in summer and drops off sharply afterward. Launch a limited-time "lock in your summer price" annual membership campaign during the June–September peak, framing membership as a way to keep riding at a lower effective cost through the season and into the following year, directly countering the seasonal drop-off that currently loses these riders.

3. Convert weekday, commute-hour casual riders with cost-comparison messaging A subset of casual riders already ride during the same 7–9 AM and 5–6 PM commute windows members do. Use in-app and email messaging that shows these specific riders a personalized cost comparison (their historical single-ride spend vs. an annual membership), positioning membership as the obvious upgrade for a commute they are already making.